QUANTUM COMPUTING → [Project Q-Community].

DAY-1

~~Note~~ 1:-

Quantum computers	classical computers
Specialized problem solvers	everyday computing
Use qubits	Use bits
qubit → 0,1 & both	bit - 0 & 1 only
Includes Superposition	Does not Include it

classical bits:- [switch on or off]

→ 0 & 1

→ Define state

→ Store one_val at an instance

Qubit:

→ works in both States

$|0\rangle \rightarrow \frac{|0\rangle+|1\rangle}{2} \quad |1\rangle$

$\downarrow$

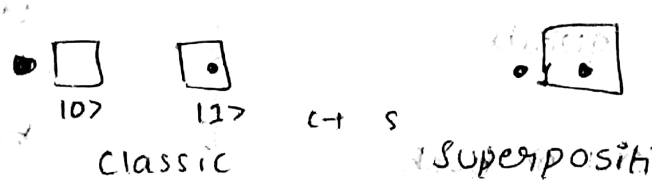
$|0.5\rangle$

→ Exists as a prob mixtures of both states until measurment

CORE PRINCIPLES INCLUDED:-

1. **Qubit:-**
   - → Fundamental unit
   - → switching of a coin
   - → exist in multiple states simultaneously

2. **Superposition:-**
   - → exists,



3. **Entanglement:-**
   - → changing info from |0> to |1>
   - → enable powerful quantum algorithms

4. **Interference:-**
   - → amplify correct answers & suppress the incorrect ones, C → D
   - → use visual representation

5. **Decoherence:-**
   - → Ideal Q - State

     ↓

     Env noise

     ↓

     Loss - Info

     ↓

     Decoherence

Date: 2/3-

Qiskit :-

- IBM → open source SDK
  - ↳ Build circuits
  - ↳ Simulate programs
  - ↳ Access real quantum computers
  - ↳ analyze results

IBM Q-stack

Dev
↓
Qiskit
↓
IBM Runtime
↓
Q - Hardware
↓
Results

Recaps:-

- Qubits
- Bits
- Core Principles
- Qiskit
- Q-computing vs classical